A Numerical Method for Controllability Problems for the Wave Equation

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1 Introduction

Let T denote a given positive number and let $u_0(\mathbf{x})$ and $u_1(\mathbf{x})$ denote given functions defined on $(0,1)$. Let $\Sigma = \{0,1\} \times (0,T)$, $Q = (0,1) \times (0,T)$ and $(u_0, u_1) \in L^2(0,1) \times H^{-1}(0,1)$. The *exact Dirichlet boundary controllability* problem for the wave equation is: *find a control function $g(\mathbf{x},t)$ defined on Σ such that u satisfies*

$$\begin{cases} u_{tt} - u_{xx} = 0 \quad \text{in } Q \\ u|_{t=0} = u_0 \quad \text{and} \quad u_t|_{t=0} = u_1 \quad \text{in } (0,1) \\ u|_{t=T} = 0 \quad \text{and} \quad u_t|_{t=T} = 0 \quad \textit{in } (0,1) \\ u = g \quad \text{on } \Sigma\,. \end{cases} \tag{1}$$

It is well known that state and control functions u and g such that (1) is satisfied exist provided T is sufficiently large; see, e.g., [7, 8, 11, 10]. In general, solutions of the Controllability problem (1), when they exist, are not unique. However, it can be shown [7, 8] that a solution of the controllability problem (1) having minimum $L^2(\Sigma)$-norm is unique.

More generally, one can consider the *exact boundary controllability problem* (1) with the boundary condition $u = g$ on Σ is replaced by $B(u) = g$ on Σ, where B denotes a boundary operator. For example, for $B(u) = (\partial_n u)|_\Sigma = g$, one has the *Neumann* exact boundary controllability problem for the wave equation. One can also consider more general criteria for extracting a particular solution of the controllability problem from the set of all solutions. For example, one can replace the minimum $L^2(\Sigma)$-norm requirement for the control function g by requiring that the solution minimize a norm-like functional $\mathcal{H}(u)$ over an appropriate function space.

A numerical method was developed in [2, 7] to determine approximations of the minimum $L^2(\Sigma)$-norm solution of (1). However, the computational experiments in [2] showed that the approximate solutions obtained in this manner do not converge. Subsequently, a Tychonoff regularization method was added to the solution procedure after which convergent approximations were obtained, at least for smooth controls. Numerical approximation of (1) were also studied in [5, 12, 9]. In these papers, the convergence of the boundary controls for finite difference discretized equations to the control of the continuous 1-D wave equation as the mesh size tends to zero has been proved if the step size in time equals to that in space. Furthermore, if the step size in time is less than that in space, regularization techniques were used to obtain convergent approximations.

Another computational approach was proposed in [4] which obtained an approximation to the minimum $L^2(Q)$-norm (not $L^2(\Sigma)$-norm) solution of (1). Here, we develop that method so that controls satisfying other criteria, including minimizing the $L^2(\Sigma)$-norm can be obtained.

2 Coupled space-time finite difference discretization

Let $\Omega = (0,1)$. Then, the minimum value of T for which exact controllability holds is 1 so that, throughout, we assume that $T \geq 1$. We employ a finite difference discretization for the wave equation. Let $x_i = i/J$, $i,j = 0,1,\ldots,J$ and $t_k = kT/K$, $k = 0,1,\ldots,K$, where J and K are chosen positive integers, denote a partition of $\Omega \times (0,T)$, and let $u_{i,k}$ denote the approximation to $u(x_i,t_k)$. The grid sizes in space and time are then given by $h = 1/J$ and $\delta = T/K$, repectively. The wave equation (1) is discretized using standard three-point difference quotients so that, with $\lambda = \delta/h$, the $u_{i,k}$'s are required to satisfy: for $i = 1,2,\ldots,J-1$ and $k = 1,2,\ldots,K-1$,

$$u_{i,k+1} - 2(1-\lambda^2)u_{i,k} - \lambda^2(u_{i+1,k} + u_{i-1,k}) + u_{i,k-1} = 0\,. \tag{2}$$

The standard CFL condition that guarantees the stability of the scheme (2) is given by $\lambda \leq 1$ which implies that $TJ/K \leq 1$. Then, since $T \geq 1$, we have $K \geq J$. The intial and terminal conditions are discretized by: for $i = 0,1,\ldots,J$,

$$u_{i,0} = u_0(x_i), \quad u_{i,1} = u_{i,0} + \delta u_1(x_i), \quad u_{i,K} = 0, \quad u_{i,K-1} = 0\,. \tag{3}$$

The unknows associated with the domains of dependence of the initial and terminal data are uniquely determined, after computing (3), by an explicit marching procedure using some of the equations in (2):

$$\begin{array}{l}
\text{for}\quad k = 1,\ldots,M \\
\quad\; k_1 = k,\;\; k_2 = K-k \\
\quad\; \text{for}\quad i = k,\ldots,J-k \\
\quad\quad\; u_{i,k_1+1} = 2(1-\lambda^2)u_{i,k_1} + \lambda^2(u_{i+1,k_1} + u_{i-1,k_1}) - u_{i,k_1-1} \\
\quad\quad\; u_{i,k_2-1} = 2(1-\lambda^2)u_{i,k_2} + \lambda^2(u_{i+1,k_2} + u_{i-1,k_2}) - u_{i,k_2+1} \\
\quad\; \text{end} \\
\text{end}
\end{array} \tag{4}$$

(Without loss of generality, we have assumed that $J = 2M+1$ for some integer $M > 0$; the case of J being an even integer can be treated in a similar manner.) The linear system formed by (2) and (3) has $p = (J -$

$1)(K-1)+4(J+1)$ equations and $q=(J+1)(K+1)$ unknowns. We symbollically express it in the form $A\mathbf{U}=\mathbf{F}$, where the matrix $A\in\mathbb{R}^{p\times q}$ has bandwidth proportional to $J+1$, $\mathbf{U}\in\mathbb{R}^q$ is a vector of the unknowns $u_{i,k}$, and $\mathbf{F}\in\mathbb{R}^p$ is a known vector determined from the intial data. It is also worth noting that the number of non-zero elements of each row or each column of A is less than or equal to 5. Furthermore, we have that $q-p=(J+1)(K-3)-(J-1)(K-1)=2(K-J)-4$ so that, if we choose $K>J+2$, then $q-p=2(K-J)-4>4-4>0$, i.e., the linear system $A\mathbf{U}=\mathbf{F}$ has more unknowns than equations. Let $\mathbf{V}$ be the vector of the unknowns determined by (3) and (4); it is easily determined that $\mathbf{V}\in\mathbb{R}^r$, where $r=4(J+1)+(J^2-1)/2$ is also the number of equations in (3) and (4) together.

Let $\widetilde{p}=p-r=(J-1)(K-1)-\frac{1}{2}(J^2-1)$ and $\widetilde{q}=q-r=(J+1)(K-3)-\frac{1}{2}(J^2-1)$. Clearly, $q-p=\widetilde{q}-\widetilde{p}$. Let $\widetilde{\mathbf{U}}\in\mathbb{R}^{\widetilde{q}}$ be the vector of $u_{i,k}$'s not belonging to $\mathbf{V}$. Then, the linear system $A\mathbf{U}=\mathbf{F}$ can be reduced to $\widetilde{A}\widetilde{\mathbf{U}}=\widetilde{\mathbf{F}}$, where $\widetilde{A}\in\mathbb{R}^{\widetilde{p}\times\widetilde{q}}$ and $\widetilde{\mathbf{F}}\in\mathbb{R}^{\widetilde{q}}$ are obtained by deleting the equations used in (3) and (4) to pre-determine $\mathbf{V}$ and then, in the remaining equations, moving the terms involving the components of $\mathbf{V}$ to the right-hand side. Note that we can re-express the linear system $\widetilde{A}\widetilde{\mathbf{U}}=\widetilde{\mathbf{F}}$ by the following recursive computations, i.e., by a marching procedure from the center of Q to the lateral surface Σ:

$$
\begin{aligned}
&\text{for}\quad i=M,\dots,1\\
&\qquad i_1=i,\, i_2=J-i\\
&\qquad \text{for}\quad k=i+1,\dots,K-i-1\\
&\qquad\qquad u_{i_1-1,k}=\tfrac{1}{\lambda^2}(u_{i_1,k+1}+u_{i_1,k-1})-2(\tfrac{1}{\lambda^2}-1)u_{i_1,k}-u_{i_1+1,k}\\
&\qquad\qquad u_{i_2+1,k}=\tfrac{1}{\lambda^2}(u_{i_2,k+1}+u_{i_2,k-1})-2(\tfrac{1}{\lambda^2}-1)u_{i_2,k}-u_{i_2-1,k}\\
&\qquad \text{end}\\
&\text{end}
\end{aligned}
\tag{5}
$$

It can be shown [6] that the matrix $\widetilde{A}$ (and therefore A as well) is of full row rank, i.e., the equations included in the linear system $\widetilde{A}\widetilde{\mathbf{U}}=\widetilde{\mathbf{F}}$ are linearly independent. Let $\mathbf{W}=\{u_{i,k}\,|\,i=,M,M+1,\;k=M+2,\dots,K-M-2\}$. Clearly, $\mathbf{W}\subset\mathbb{R}^s$, where $s=2(K-@M-3)$; also, the solution of the system $\widetilde{A}\widetilde{\mathbf{U}}=\widetilde{\mathbf{F}}$ can be uniquely from $\mathbf{W}$ by the marching procedure (5).

We now turn to the selection of a particular solution of the linear system $A\mathbf{U}=\mathbf{F}$, i.e., of the discretized wave equation and initial and terminal conditions, by requiring that solution to also solve a discrete optimization problem. We first focus on finding approximations of the minimum $L^2(\Sigma)$-norm solution of the controllability problem. Specifically, we use the trapezoidal rule to obtain the approximation

$$\begin{aligned}\mathcal{J}(u) \equiv ||g||^2_{L^2(\Sigma)} &= \int_\Sigma u^2 \, d\Gamma dt = \int_0^T \left[u^2(0,t) + u^2(1,t)\right] \, dt \\ &\approx \delta \sum_{k=0}^{K} \alpha_k \left({u_{0,k}}^2 + {u_{J,k}}^2\right) \equiv \mathcal{J}_h(\mathbf{U}) \,,\end{aligned} \tag{6}$$

where $\alpha_0 = \alpha_K = 1/2$ and $\alpha_1 = \cdots = \alpha_{K-1} = 1$. Furthermore, we effect the partition $\widetilde{\mathbf{U}} = (\widetilde{\mathbf{U}}_1^T, \widetilde{\mathbf{U}}_2^T)^T$, where $\widetilde{\mathbf{U}}_1$ denotes the subvector of $\widetilde{\mathbf{U}}$ belonging to the boundary $\overline{\Sigma}$ and $\widetilde{\mathbf{U}}_2$ denotes the subvector of $\widetilde{\mathbf{U}}$ belonging to the interior Q. It is seen that $\widetilde{\mathbf{U}}_1 \in \mathbb{R}^{\widetilde{q}_1}$, where $\widetilde{q}_1 = 2(K-3)$, and $\widetilde{\mathbf{U}}_2 \in \mathbb{R}^{\widetilde{q}_2}$, where $\widetilde{q}_2 = \widetilde{q} - 2(K-3)$. We are then led to the following *discrete* quadratic optimization problem:

$$\min_{\widetilde{\mathbf{U}} \in \mathbb{R}^{\widetilde{q}}} \mathcal{J}_h(\widetilde{\mathbf{U}}) = \delta(\widetilde{\mathbf{U}}_1^T \widetilde{\mathbf{U}}_1 + b) \quad \text{subject to} \quad \widetilde{A}_1 \widetilde{\mathbf{U}}_1 + \widetilde{A}_2 \widetilde{\mathbf{U}}_2 = \widetilde{\mathbf{F}} \,, \tag{7}$$

where $\widetilde{A}_1 \in \mathbb{R}^{\widetilde{p} \times \widetilde{q}_1}$ and $\widetilde{A}_2 \in \mathbb{R}^{\widetilde{p} \times \widetilde{q}_2}$ are obained by appropriately rearranging and partitioning $\widetilde{A}$ corresponding to $\widetilde{\mathbf{U}}_1$ and $\widetilde{\mathbf{U}}_2$, and b is a constant depending on the components of $\mathbf{U}$ belonging to $\overline{\Sigma}$ that are pre-determined by (3). It should be pointed out that the problem (7) couples space and time together. Regarding the solution of the optimization system (7), we have the following result.

If u denotes the minimum $L^2(\Sigma)$-norm solution of (1) and u^h the corresponding discrete solution of (6) with $\delta = h$, i.e., $\lambda = 1$, then it is shown in [9] $g^h \to g$ strongly in $L^2(\Sigma)$, where $g = u|_\Sigma$ and $g^h = u^h|_\Sigma$.

We also consider finding the soultion of the exact controllability problem (1) that also has minimum $H^1(\Sigma)$-norm which results in more regular controls g. In this case, we use

$$\begin{aligned}\mathcal{K}(u) &\equiv ||g||^2_{H^1(\Sigma)} = \int_0^T (u^2|_{(0,t)} + u_t^2|_{(0,t)} + u^2|_{(1,t)} + u_t^2|_{(1,t)}) dt \approx \mathcal{K}_h(\mathbf{U}) \\ &\approx \delta \sum_{k=0}^{K} \alpha_k \left(u_{0,k}^2 + u_{J,k}^2\right) + \frac{1}{\delta} \sum_{k=1}^{K} \Big((u_{0,k} - u_{0,k-1})^2 + (u_{J,k} - u_{J,k-1})^2 \Big)\end{aligned}$$

and are led to another discrete quadratic optimization problem:

$$\min_{\widetilde{\mathbf{U}} \in \mathbb{R}^{\widetilde{q}}} \mathcal{K}_h(\widetilde{\mathbf{U}}) = \delta(\widetilde{\mathbf{U}}_1^T R \widetilde{\mathbf{U}}_1 + b) \quad \text{subject to} \quad \widetilde{A}_1 \widetilde{\mathbf{U}}_1 + \widetilde{A}_2 \widetilde{\mathbf{U}}_2 = \widetilde{\mathbf{F}} \,, \tag{8}$$

where $R \in \mathbb{R}^{\widetilde{q}_1 \times \widetilde{q}_1}$ is a tridiagonal, positive definite matrix. This discrete problem also has a unique solution; in fact, one may view (7) as the special case of (8) for which R is the identity matrix. If $\lambda = 1$, then again the convergence of the discrete control holds.

A Lagrange multiplier method. If $\widetilde{\mathbf{U}}$ is the solution of the optimization problems (7) or (8), then the Lagrange multiplier rule states that $\widetilde{\mathbf{U}}$ is a stationary point of the Lagrangian functional $\mathcal{L}(\widetilde{\mathbf{U}}) = \delta(\widetilde{\mathbf{U}}_1^T R \widetilde{\mathbf{U}}_1 + b) - \mathbf{S}^T(\widetilde{A}_1 \widetilde{\mathbf{U}}_1 +$

$\widetilde{A}_2\widetilde{\mathbf{U}}_2 - \widetilde{\mathbf{F}})$, where $\mathbf{S} \in \mathbb{R}^{\widetilde{p}}$ is the Lagrange multiplier introduced to enforce the constraint equations $\widetilde{A}_1\widetilde{\mathbf{U}}_1 + \widetilde{A}_2\widetilde{\mathbf{U}}_2 = \widetilde{\mathbf{F}}$ and $R = I$ in case of (7). Then, it is easily shown that $\widetilde{\mathbf{U}} = (\widetilde{\mathbf{U}}_1, \widetilde{\mathbf{U}}_2)$ and $\widetilde{\mathbf{S}} = \mathbf{S}/\delta$ are the solution of

$$\begin{pmatrix} 0 & \widetilde{A}_2^T \\ \widetilde{A}_2 & -\widetilde{A}_1^T R^{-1}\widetilde{A}_1 \end{pmatrix} \begin{pmatrix} \widetilde{\mathbf{U}}_2 \\ \widetilde{\mathbf{S}} \end{pmatrix} = \begin{pmatrix} 0 \\ \widetilde{\mathbf{F}} \end{pmatrix} \qquad \text{and} \qquad \widetilde{\mathbf{U}}_1 = -R^{-1}\widetilde{A}_1^T\widetilde{\mathbf{S}}\,. \tag{9}$$

Note that $\widetilde{\mathbf{U}}_1$ along with the boundary $u_{i,k}$'s found from (3) and (4) can be used to define the approximation to the exact Dirichlet boundary control $u|_\Sigma = g$. Clearly, the coefficient matrix of the linear system in the first equation in (9) is symmetric but it is not positive definite. Note that $\widetilde{A}_1$ and $\widetilde{A}_2$ are sparse matrices having, like A, at most 5 non-zero elements in each of their rows or columns. Thus, it is efficient to use iterative methods, e.g., BiCG, SYMMQL, or CGNR, to solve the first equation in (9). In higher dimensions, the application of a preconditioner may be useful since the dimension of the the first equation in (9) may be large. In a practical implementation, the matrices $\widetilde{A}_1$ and $\widetilde{A}_2$ need not be stored. It is also worth noting that, at each step of an iteration, we may only need to update $\widetilde{\mathbf{S}}$ and that part of $\mathbf{W}$ included in $\widetilde{\mathbf{U}}_2$; then the remainder of $\widetilde{\mathbf{U}}_2$ can be directly computed by the marching procedure (5).

Reduction to an unconstrained optimization problem. Another method for solving the optimization problems (7) or (8) uses the marching procedure (5) to determine $\widetilde{\mathbf{U}}_1$ from $\mathbf{W}$, provided that $\mathbf{V}$ is pre-calculated using (3) and (4). Thus, $\widetilde{\mathbf{U}}_1$ is a linear function of $\mathbf{W}$, i.e., there exists a matrix $D \in \mathbb{R}^{\widetilde{q}_1 \times s}$ such that $\widetilde{\mathbf{U}}_1 = D\mathbf{W} + \mathbf{C}$, where $\mathbf{C} \in \mathbb{R}^{\widetilde{q}_1}$ is determined from $\mathbf{V}$. Then, the optimization problem (8) (or (7) if we set $R = I$) is equivalent to $\min_{\mathbf{W}\in\mathbb{R}^r} \mathcal{K}_h(\widetilde{\mathbf{U}}_1)$ subject to $\widetilde{\mathbf{U}}_1 = D\mathbf{W} + \mathbf{C}$. Substituting $\widetilde{\mathbf{U}}_1 = D\mathbf{W} + \mathbf{C}$ into $\mathcal{K}_h(\widetilde{\mathbf{U}}_1)$, we obtain $\mathcal{K}_h(\widetilde{\mathbf{U}}_1) = \widetilde{\mathcal{K}}_h(\mathbf{W}) = \mathbf{W}^T(D^TRD)\mathbf{W} + 2\mathbf{C}^TRD\mathbf{W} + \mathbf{C}^TR\mathbf{C}$ so that the constrained optimization problem for $\mathcal{K}_h(\widetilde{\mathbf{U}}_1)$ is equivalent to the *one-dimensional unconstrained* optimization problem $\min_{\mathbf{W}\in\mathbb{R}^r} \widetilde{\mathcal{K}}_h(\mathbf{W})$. Then, $\mathbf{W}$ is a stationary point of $\widetilde{\mathcal{K}}_h(\cdot)$, we obtain $D^TRD\mathbf{W} = -2D^TR\mathbf{C}$. Since D^TRD is a positive definite matrix, we can use, e.g., the conjugate gradient method, to solve this linear system.

The matrix $D \in \mathbb{R}^{\widetilde{q}_1 \times s}$ is a full matrix. In higher dimensions, $\widetilde{q}_1$ is a large number even for small J and K so that directly storing D requires large amounts of memory. However, recall that D embodies the marching procedure (5); that procedure is done level by level from the center of the space-time rectangle Q to its lateral sides Σ. In fact, the marching procedure can be expressed as

$$\begin{cases} \mathbf{U}_{M-1} = D_{3,M-1}\mathbf{W} + \mathbf{C}_{M-1} \\ \mathbf{U}_{M-2} = D_{2,M-2}\mathbf{U}_{M-1} + D_{3,M-2}\mathbf{W} + \mathbf{C}_{M-2} \\ \mathbf{U}_m = D_{1,m}\mathbf{U}_{m+1} + D_{2,m}\mathbf{U}_{m+2} + \mathbf{C}_m \quad \text{for} \quad m = M-3, \ldots, 0 \end{cases}$$

where $\mathbf{U}_m = \{u_{i,k} \in \widetilde{\mathbf{U}} \,|\, i = m \,\text{or}\, i = J - m\}$ for $m = 0, 1, \ldots, M-1$ and the $\{D_{i,m}\}$ are matrices. Clearly, $\widetilde{\mathbf{U}}_1 = \mathbf{U}_0$ and each $D_{i,m}$ is a bounded matrix having at most 5 non-zero elements in each of its rows or columns and D can be represented by the sequence of matrices $\{D_{i,m}\}$ corresponding to each level of the marching procedure (5). Consequently, we need only save $\{D_{i,m}\}$ to completely determine the action of D and D^T on a vector.

Note that this implementation cannot be directly applied to implicit (in time) approximations of the wave equation or to finite element approximations involving (unlumped) mass matrices. However, its extension to higher dimensions is straightforward.

3 Computational experiments

An example with known smooth exact solution, A smooth exact solution to the minimum $L^2(\Sigma)$-norm controllability problem can be constructed by using Fourier series in a similar way to that used in [2]. Suppose that $Q = (0,1) \times (0, 7/4)$ and $\Sigma = \{0,1\} \times (0, 7/4)$. Let $\psi_0(x,t) = -\sqrt{2}\pi \cos \pi(t - \frac{1}{4}) \cos 2\pi x$ and

$$\begin{aligned}\psi_1(x,t) &= \left[2\sqrt{2}\pi(T-t)\sin(t-\frac{1}{4}) - \frac{10}{3}\sqrt{2}\sin\pi(t-T)\right]\sin\pi x \\ &+ \sum_{p\geq 3 \text{ and } p \text{ odd}} \frac{4\sqrt{2}}{p^2-1}\left[\frac{3p}{p^2-4}\cos\pi(t-\frac{1}{4}) + \sin p\pi(t-T)\right]\sin p\pi x\,.\end{aligned}$$

Then, set the initial conditions $u_0(x) = \psi_0(x,0) + \psi_1(x,0)$ and $u_1(x) = \partial\psi_0/\partial t(x,0) + \partial\psi_1/\partial t(x,0)$. It is worth noting that u_0 is a Lipschitz continuous function but does not belong to $C^1(\overline{\Omega})$ and u_1 is a bounded function but does not belong to $C^0(\overline{\Omega})$. For this initial data, it can be shown that $u(x,t) = \psi_0(x,t) + \psi_1(x,t)$ is the exact solution having minimum $L^2(\Sigma)$-norm of the controllability problem (1). The corresponding exact Dirichlet control g is then given by restricting $u(x,t)$ to the lateral sides Σ, i.e., $g(t) = u(0,t) = u(1,t) = -\sqrt{2}\pi\cos\pi(t - \frac{1}{4})$. For future reference, note that $\|g\|_{L^2(\Sigma)} \approx 6.1340$ and $\|u\|_{L^2(Q)} \approx 5.8656$.

We apply our numerical method to this example. Computational experiments were carried out for $h = 1/16$, $1/32$, $1/64$, and $1/128$ with $\lambda = 1$ and $\lambda = 7/8$ repectively, so that the stability condition $K \geq J$ is satisfied. The results of our computational experiments are summarized in Table 1, where u^h and g^h are the computed approximations of the exact solutions u and g, respectively. From this table, it seems that g^h converges to g in the $L^2(\Sigma)$-norm at a rate of roughly 1 as does u^h in the $L^2(Q)$-norm. In order to visualize the convergence of our method as h becomes smaller, in Figure 1 we provide plots of the exact solution g and the corresponding computed discrete solutions g^h for 1/64 and 1/128. It seems that our method produces (pointwise) convergent approximations for both $\lambda = 1$ and $\lambda = 7/8$ *without the need for regularization.* This should be contrasted with other methods, e.g., see [2], for which for $\lambda < 1$, regularization was needed in order to obtain convergence. It may at first seem that our results for $\lambda < 1$ are contradictory to the result of [9] which states that there exists a sequence of approximate controls that do not converge. However, that result says nothing about the *particular* sequence of minimum discrete $L^2(\Sigma)$-norm controls that is determined by our method.

Table 1. Results of computational experiments for the minimum $L^2(\Sigma)$-norm case for the example having smooth solutions for the control.

h		1/16	1/32	1/64	1/128
$\|g^h\|_{L^2(\Sigma)}$	$\lambda = 1$	5.9572	6.0313	6.0804	6.1008
	$\lambda = 7/8$	5.9803	6.0507	6.0890	6.1104
$\|g^h - g\|_{L^2(\Sigma)}/\|g\|_{L^2(\Sigma)}$	$\lambda = 1$	10.858%	4.625%	2.071%	1.051%
	$\lambda = 7/8$	9.662%	5.003%	2.137%	0.991%
$\|u^h\|_{L^2(Q)}$	$\lambda = 1$	5.6954	5.7873	5.8266	5.8442
	$\lambda = 7/8$	5.7249	5.7899	5.8327	5.8470
$\|u^h - u\|_{L^2(Q)}/\|u\|_{L^2(Q)}$	$\lambda = 1$	9.283%	4.153%	1.989%	1.012%
	$\lambda = 7/8$	7.636%	3.907%	1.864%	0.891%

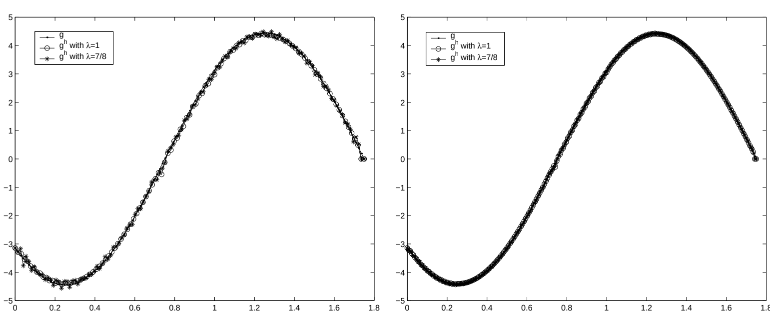

Fig. 1. The exact minimum $L^2(\Sigma)$-norm Dirichlet control g and the approximate controls g^h vs. time along $x = 0$ (and $x = 1$) for the example having smooth solutions for the control. Left: $h = 1/64$; right: $h = 1/128$.

Generic example with minimum $L^2(\Sigma)$-norm and minimum $H^1(\Sigma)$-norm boundary controls. In the example discussed above, the minimum $L^2(\Sigma)$-norm control is very smooth. Using our methods, we obtained good approximations for this example without the need for regularization. However, this is not the generic case. In general, even for smooth initial data, the minimum $L^2(\Sigma)$-norm Dirichlet control for the controllability problem (1) will not be smooth. In this section, we illustrate this point and also examine the performance of our method for the generic case.

We again choose $Q = (0,1) \times (0,7/4)$ and consider the following $C^\infty(\overline{\Omega})$ initial data: $u_0(x) = \sin(\pi x)$ and $u_1(x) = \pi \sin(\pi x)$. Note that the initial conditions vanish at the boundary and, that due to symmetry, we have that $u(0,t) = u(1,t)$, i.e., the control at the two sides of Q are the same.

We look at the case $\lambda = 1$. In Figure 2, we show the results for the control for several grid sizes ranging from $h = 1/32$ to $h = 1/256$. The (pointwise) convergence of the approximations is evident. Note that for these initial conditions, the minimum $L^2(\Sigma)$-norm control is seemingly piecewise smooth, i.e., it contains jump discontinuities. The pointwise convergence of the approximate control for the case of $\lambda = 1$ is probably a one-dimensional artifact; it is likely due to the fact that both the space and time variables in the wave equation in one dimension can act as time-like variables. are seemingly first order. Since the exact solutions corresponding to these initial conditions are not known, we will take the seemingly converged solutions obtained with $h = 1/256$ and $\lambda = 1$ as a baseline to which we compare our other computaional results. We will denote this baseline approximate control by g^*.

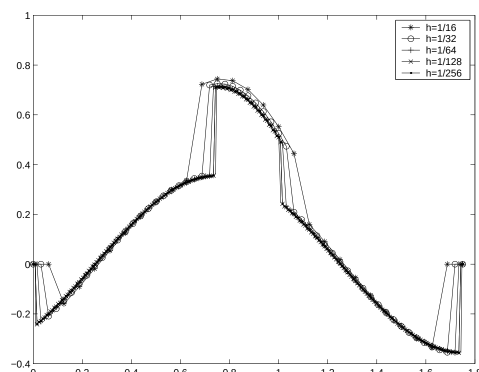

Fig. 2. The approximate minimum $L^2(\Sigma)$-norm Dirichlet control g^h vs. time for $\lambda = 1$ forthe example with piecewise smooth solutions at $x = 0$ (and $x = 1$).

Computational experiments were also carried out for $\lambda = 7/8$ and $\lambda = 7/12$ for several values of the grid size ranging from $h = 1/32$ to $h = 1/128$. The results are summarized in Table 2. In Figure 3, we provide plots of the computed discrete solution g^h for the two different values of λ and for different values of the grid size. From Figure 3, we see that the approximate minimum $L^2(\Sigma)$-norm Dirichlet controls obtained with values of $\lambda < 1$ are highly oscillatory. In fact, the frequencies of oscillation increase with decreasing grid size. However, is seems that the amplituded of the oscillationd do not increase as the grid size decreases. Furthermore, from the results in Table 2, is seems that for $\lambda < 1$, the approximate controls g^h, although oscillatory in a nature and not convergent in a pointwise sense, converge in an $L^2(\Sigma)$ sense. Also, the solution u^h is seemingly convergent in an $L^2(Q)$ sense. The results of Table 2 and Figure 3 indicate that for the generic case of non-smooth minimum $L^2(\Sigma)$ controls and for general $\lambda < 1$, our method produces convergent (in $L^2(Q)$ and $L^2(\Sigma)$) approximations *without the need of regulatization* but the approximations are not in general convergent in a pointwise sense. Of course, approximations that do not converge in a pointwise sense may be of no practical use, even if they converge in a root mean square sense.

Table 2. Results of computational experiments for the minimum $L^2(\Sigma)$-norm case for the example with piecewise smooth solutions and for $\lambda < 1$.

	λ	$h = 1/16$	$h = 1/32$	$h = 1/64$	$h = 1/128$
$\Vert g^h \Vert_{L^2(\Sigma)}$	7/8	0.6715	0.6354	0.6147	0.6036
	7/12	0.6550	0.6212	0.6085	0.5977
$\Vert u^h \Vert_{L^2(Q)}$	7/8	0.7546	0.7327	0.7204	0.7140
	7/12	0.7431	0.7234	0.7164	0.7124

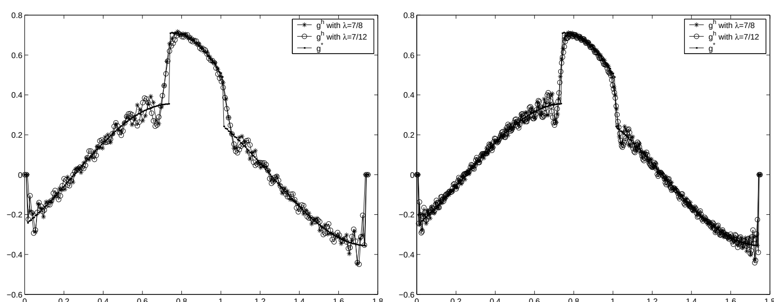

Fig. 3. The approximate minimum $L^2(\Sigma)$-norm Dirichlet controls g^h vs. time at $x = 0$ (and $x = 1$) for the example with piecewise smooth solutions. Left: $h = 1/64$; right: $h = 1/128$.

So far, the computational experiments indicate that, generically, minimum $L^2(\Sigma)$-norm boundary controls are only piecewise smooth and that approximations obtained by our method are in general not pointwise convergent so that they are not be realizable in practice. In order to obtain smoother controls, i.e., at least continuous in time, we consider the particular solution of the exact controllability problem (1) having minimum $H^1(\Sigma)$-norm. We use the same initial data as for Figure 3.

Figure 4 provides results of computational experiments for minimum $H^1(\Sigma)$-norm controls. It indicates that, in general, minimum $H^1(\Sigma)$-norm solutions of the controllability problem are smooth and that the approximate controls obtained by our method converge in a pointwise sense for both $\lambda = 1$ and $\lambda < 1$ without the need for regularization. Our results also indicate that our method produces convergent (in $L^2(Q)$ and $L^2(\Sigma)$) approximations.

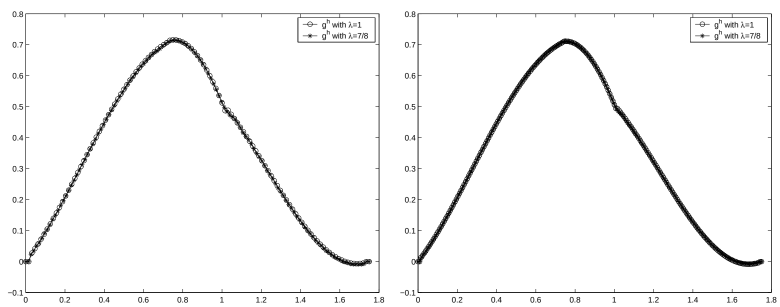

Fig. 4. The approximate minimum $H^1(\Sigma)$-norm Dirichlet controls g^h vs. time at $x = 0$ (and $x = 1$). Left: $h = 1/64$; right: $h = 1/128$.

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